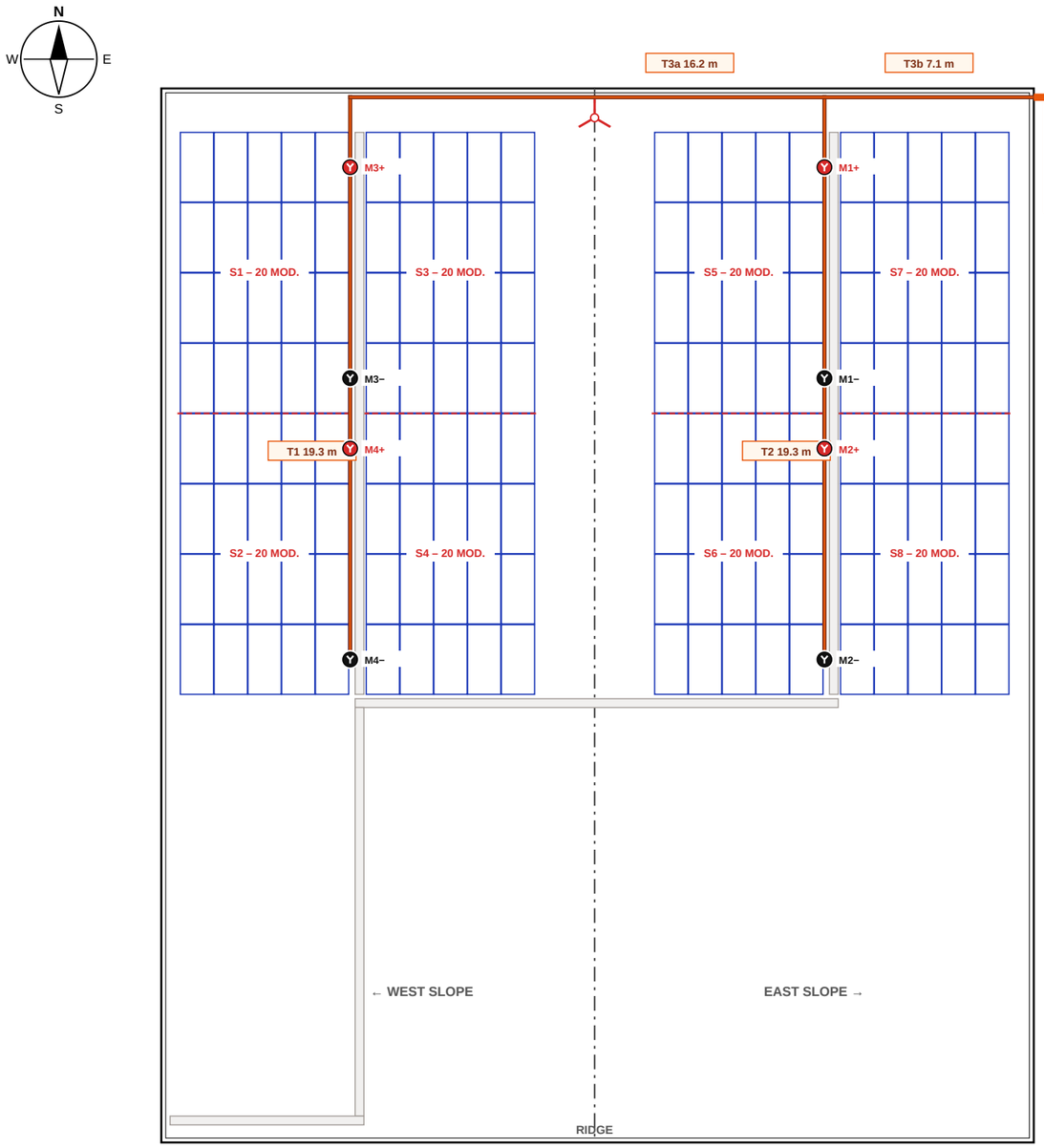
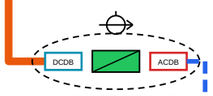




INV. 80 kW + DCDB + ACDB
(EAST WALL, GROUND LEVEL)

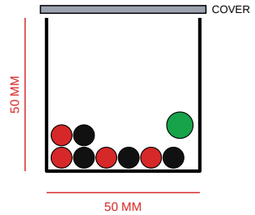
T4 13.9 m
DOWN WALL ≈ 10 m
+ ALONG WALL

AC: 95 SQ MM
AL ARMOURED
TO LT PANEL
≈ 20 m *
150 x 50 MM
TRAY / TRENCH

CABLE TRAY SCHEDULE (DC)

ID	ROUTE	LENGTH	CABLES IN TRAY
T1	GAP BLOCK 1/2: ROW 8 → NORTH EDGE	19.3 m	4 x 6 SQ MM + 16 SQ MM E (MPPT 3, 4)
T2	GAP BLOCK 3/4: ROW 8 → NORTH EDGE	19.3 m	4 x 6 SQ MM + 16 SQ MM E (MPPT 1, 2)
T3a	NORTH EDGE: T1 → T2 (OVER RIDGE)	16.2 m	4 x 6 SQ MM + 16 SQ MM E (MPPT 3, 4)
T3b	NORTH EDGE: T2 → EAST EAVE	7.1 m	8 x 6 SQ MM + 16 SQ MM E (ALL 4 MPPT)
T4	EAST WALL: DOWN 10 m + ALONG TO INVERTER	13.9 m	8 x 6 SQ MM + 16 SQ MM E (ALL 4 MPPT)
	TOTAL DC TRAY 50 x 50 MM	75.9 m	
	ORDER (+10 %), 2.5 m LENGTHS	83 m	34 LENGTHS + COVERS

TRAY SECTION (T3b / T4, FULL LOAD)



8 x 6 SQ MM DC (Ø ≈ 6.8 MM; RED +VE, BLACK –VE)
+ 1 x 16 SQ MM Cu EARTH (GREEN, Ø ≈ 8.6 MM)
FILL ≈ 349 / 2,500 MM² ≈ 14 % (LIMIT 40 %)
WITH Y-CONNECTORS THE TRAY CARRIES HALF THE CORES
OF THE NO-Y DESIGN — 50 x 50 IS ENOUGH.

CABLE TRAY NOTES

1. 50 x 50 MM PERFORATED HDG TRAY, 1.6 MM, WITH BOLTED COVER (UV & BIRD / MONKEY PROTECTION).
2. ONE SIZE THROUGHOUT, FITTINGS COMMON.
3. RAISE TRAY 75–100 MM ABOVE SHEET ON PURLIN CLAMPS EVERY 1.5 m; WALL BRACKETS EVERY 1.2 m.
4. FITTINGS: 1 FLAT BEND, 1 TEE, 2 VERTICAL BENDS, COUPLERS AT EVERY JOINT, 2 END CAPS.
5. Y-CONNECTORS STAY OUTSIDE THE TRAY, CLIPPED TO THE RAIL BESIDE IT, NEVER LYING IN WATER.
6. BOND TRAY SECTIONS TOGETHER AND TO EP-1.
7. AC CABLE IN ITS OWN TRAY / TRENCH, ≥ 300 MM FROM DC.
8. LA DOWN CONDUCTOR NEVER IN OR ON THE TRAY.

DC HOME-RUN CABLE LENGTHS (ONE +/– PAIR PER MPPT, 6 SQ MM)

MPPT	STRINGS (Y)	SLOPE	Y POSITION	+VE 6 SQ MM	–VE 6 SQ MM	LOOP	ΔV @ 30.1 A
MPPT 1	S5 + S7	EAST	GAP 3/4, ROW 1 / 4	25.4 m	32.6 m	58.1 m	0.71 %
MPPT 2	S6 + S8	EAST	GAP 3/4, ROW 5 / 8	35.0 m	42.3 m	77.3 m	0.95 %
MPPT 3	S1 + S3	WEST	GAP 1/2, ROW 1 / 4	41.6 m	48.8 m	90.5 m	1.11 %
MPPT 4	S2 + S4	WEST	GAP 1/2, ROW 5 / 8	51.2 m	58.5 m	109.7 m	1.34 %
TOTAL				153.4 m	182.2 m	335.6 m	

EACH RUN = Y → TRAY NORTH → NORTH EDGE → DOWN EAST WALL (10 m) → INVERTER, + 0.5 m SLACK AT THE Y + 1.5 m AT THE INVERTER.
ΔV = 30.1 A x 3.39 mΩ/m x LOOP ÷ 832 V (STC). 4 SQ MM STRING-LEVEL LEADS ADD < 0.02 %.

DC CABLE & CONNECTOR QUANTITIES

ITEM	SPEC	QTY	ORDER
HOME RUNS	6 SQ MM SOLAR DC (RED + / BLACK –)	336 m	355 m
FAR-STRING LEADS	4 SQ MM, 8 x 1.5 m (S3, S4, S7, S8 ±)	12 m	
ROW-TURN JUMPERS	4 SQ MM, 24 x 2.5 m (3 / STRING)	60 m	
4 SQ MM TOTAL	LEADS + JUMPERS	72 m	80 m
Y-CONNECTOR SET	1 x + (M-M-F) & 1 x – (F-F-M), ≥ 40 A	4 SETS	4 + 1 SPARE
MC4 PAIRS	HOME RUNS 8 + LEADS 8 + JUMPERS 24	40	44

CABLE ≈ ₹34,460 (355 m x ₹84 + 80 m x ₹58).

NOTE: TRAYS T1 / T2 SIT IN THE ≈ 0.6 m GAPS BESIDE WALKWAYS W1 / W2; THE Y-CONNECTORS (◉, SEE L-01) SIT BESIDE THEM AT ROWS 1, 4, 5 AND 8. TRAYS RUN NORTH, WALKWAY CONNECTOR RUNS SOUTH — NO CROSSINGS. GAPS AND ROOF SIZE SCALED FROM THE LAYOUT IMAGE (*) — VERIFY ON SITE AND RE-CUT THE HOME RUNS TO THE MEASURED ROUTE.
LA DOWN CONDUCTOR CROSSES T3a ONCE, AT 90°, RAISED ≥ 0.5 m ON AN INSULATED STAND-OFF, THEN RUNS ≥ 1 m NORTH OF THE TRAY (SEE CEIG SHEET 03).

LEGENDS

	SOLAR MODULES
	DC CABLE TRAY 50 x 50 MM
Y Y	Y-CONNECTOR + / – (M = MPPT)
	AC CABLE ROUTE (95 SQ MM AL)
	WALKWAY (SEE L-02)
	LIGHTNING ARRESTER
	STRING INVERTER

SYSTEM DESCRIPTION

SITE LOCATION	— (LAT, LONG TO BE ADDED)
DC CAPACITY	100.00 kWp
TOTAL PLANT CAPACITY (AC)	80 kW
DC / AC RATIO	1.25
MODULE CAPACITY	625 Wp (ADANI N-TOPCon)
MODULE DIMENSION (L X W X H)	2382 X 1134 X 30 mm *
TOTAL PV MODULES	160 Nos.
INVERTER CAPACITY	80 kW (GOODWE GW80K-MT) 1 No.
AZIMUTH / TILT	EAST-WEST (90° / 270°), ROOF SLOPE

CUSTOMER SIGN AND STAMP:

REV.	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	APPROVED BY
R0	29-10-26	FIRST ISSUE			

CLIENT:	**JAIN GRANITE & MARBLE**
EPC CONTRACTOR:	**NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara
DESIGN BY:	**NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara

DRAWING NO.: L-03

DRAWING NAME: **CABLE TRAY + DC CABLE LENGTHS (Y)**

PROJECT TITLE: 100.00 kWp (DC)_ROOFTOP SOLAR PLANT

RELEASED FOR: ☐ PRELIMINARY ☐ TENDER ☐ REVISION ☐ FABRICATION ☐ CONSTRUCTION ■ APPROVAL

SCALE: **1:200 (A3)**

DESIGN BY	DRAWN BY	CHECKED BY	APPROVED BY	PAPER SIZE	SHEET NO.
29-10-26	29-10-26	29-10-26	29-10-26	**A3**	**03 OF 03**